

Predicting Injury Risk of Professional Footballers

Njål Svensen & Sander Bakke

December 18, 2024

Problem motivation

We want to create a tool to help football clubs manage the fitness of their players. Clubs spend a great amount of resources to keep their players fit, as injuries can have huge personal and collective consequences. Currently, there is a big discussion about the amount of competitive games clubs (and therefore their players) are forced to play [Tra24]. Therefore, the demand for injury risk management is more valuable than ever. The model would hopefully help determine which factors contribute to injury as well, so clubs can adjust to this.

Method

Player injury is a very complex issue, and therefore has many possible solutions. The task could be treated as either a classification problem or a regression problem. The most practical and useful approach would be to define the output, “risk of getting an injury in the next time period”, as a value between 0 and 1, used to decide if a player is overworked or not. From there, if a player exceeds a given threshold, they could be classified as “going to get injured”. To achieve this, possible machine learning algorithms would be logical regression models, and for better results random forest regression and gradient boost algorithms.

Features

There are almost unlimited features that contribute to the risk of injury. Some are very hard to obtain, like training load, certain medical information of players, mental state of players etc., player data the club wants to keep secret. However there are some definitive features that contribute to injury risk we can get our hands on.

- Age, Height, Weight
- minutes played, match stats (intensity), position on the pitch
- Previous injury record
- Travel related to match schedules, weather data

Experiments

Evaluation

A challenge with this problem is the evaluation of the model performance. Since “injury risk” is not a measurable value, and certainly cannot be compared to “real” injury risk, validating the results with the typical regression evaluation metrics (MSE, RMSE, MAE, R-Score) will be difficult. As previously mentioned, the project can easily be changed into a binary classification case of “at risk” or “not at risk”, by setting a threshold or evaluating “going to be injured in the next time period”. Then F1-score, precision and recall can be used to evaluate.

Multiple different models will be used to better the results. Comparing these will also be a part of the evaluation.

Feature Analysis

Feature analysis will be performed to determine which factors contribute to injury. By obtaining this insight, further optimization can be achieved by for example omitting bad features, or finding different ways of representing certain features. In particular “match load” and “previous injury” can be represented in many different ways.

Datasets

- **Transfermarkt:** This website provides player profiles, injury histories, and statistics like age, height, and weight, as well as past injuries.
- **FBref:** This source contains comprehensive player performance data such as minutes played, goals, assists, and other match statistics that can be useful for understanding players’ physical load and position on the pitch.
- **Football-Data.co.uk:** This dataset includes historical match data, including travel schedules and match schedules, which will be used to calculate the impact of travel on injury risk.

We recognize that gathering a comprehensive and complete dataset will be the biggest challenge with this task. The model should use a dataset containing injury information as well as the other features. This probably requires a lot of pre-processing and feature engineering, which as stated in the project description, could take a considerable amount of time. Our hope is to make a complete CSV file and then utilize **pandas** to do data cleaning and set up the machine learning model.

Prior research

As the usefulness of a model, as described in this document, is very high, much prior research has been conducted. These studies had greater access to relevant data than we will have.

One such study is “Machine Learning for Understanding and Predicting Injuries in Football” [Ope22]. Where machine learning was applied to predict player injuries based on training load, player attributes and previous injuries.

References

- [Ope22] Sports Medicine Open. Machine learning for understanding and predicting injuries in football. *Sports Medicine Open*, 8(1):Article 12, 2022.
- [Tra24] Transfermarkt. Rodri suggests potential player strike - but how overworked are footballers in the modern game? *Transfermarkt*, 2024.